Microeconomic Foundations, I:
Choice and Competitive Markets

Instructor's Manual

Chapter 14: General Equilibrium

Notes on the Chapter

This chapter begins a three-chapter unit on General Equilibrium. It provides basic definitions and results, existence of equilibrium, and brief discussion of other mathematical issues connected to Walrasian equilibrium (essentially, what can be said about the set of equilibria). Chapter 15 concerns efficiency of equilibria and the core, and Chapter 16 is about time and uncertainty and general equilibrium.

There isn't a lot to say about the definitions or basic results; it's about what you would expect, including a substantial exposition of the Edgeworth Box. On existence, I provide a bit of a sampler. First, I give full details on the generalized-game style of proof. Then somewhat more briefly, I discuss proofs that begin with assumptions on aggregate excess demand; I give a relatively easy version of the Debreu–Gale–Kuhn–Nikaido Lemma (where aggregate demand is defined and bounded on the entire unit price simplex, including the boundaries) and then a version of Hildenbrand–Neuefeind, where excess demand is defined only for strictly positive prices, is bounded below, and diverges as prices approach the boundary of the simplex. In each case, I use Kakutani, although Problem 14.8 (solved in the *Guide*) pushes the student to a proof for excesss demand functions defined on the full price simplex, using Brouwer.

(Concerning fixed-point theorems, Appendix 8 provides the basics. The chapter uses Kakutani exclusively, although Problem 14.8 will acquaint the student with Brouwer, which is discussed in the appendix. I also discuss the Contraction Mapping Theorem and Tarski's Fixed-Point Theorem in the appendix, just on general principles. I had a hard time deciding, but in the end left out Eilenberg–Montgomery, to avoid having to explain about absolute retracts. Any student who is going to employ Eilenberg–Montgomery [or, for that matter, anything fancier than what I have in the appendix] would almost surely do better to consult a serious source.)

Going back to the chapter, I do very little on the structure of the set of equilibria; this is no more than a brief discussion of what is out there in the literature. I do state Mas-Colell's (1977) Theorem (using an extension of the Sonnenschein–Mantel–Debreu Theorem) that "anything goes," but I don't prove it. (Problem 14.10, which is quite easy, gives the excess-demand function that works; what is missing is Mas-Colell's extension of the S–M–D Theorem that, in any case, I didn't prove in Chapter 13.) And both on conditions for unique equilibrium and that equilibria are finite in number and isolated for generic economies, I merely point out that there are results out there and point the reader to the excellent treatment of these issues in Mas-Colell, Whinston, and Green (MWG).

My reason for leaving this material out is two-fold: (1) I tried to write a book that could be covered (at least, by me, at Stanford) in twenty ninety-five-minute sessions over ten weeks, and I am already over budget by somewhere between one and two chapters. While I am certain there will be disagreements about this, these are topics that I deemed to be of lesser importance than other material I have included. Of course, the counter argument is that I could include this and leave it up to the instructor what material to cover. But I did want to avoid a "kitchen sink" approach. And, perhaps even more, (2) I see no way to do more than repeat the excellent treatment of these issues in MWG, short of providing a course in differential geometry and topology. Not wishing to do that, not being able to add anything to MWG, and in any event wanting to encourage serious students to have more than one book on their shelves for material this fundamental, I decided just to point the reader to their treatment.

To cover fully all the material in this chapter in one class session is beyond my abilities, and I imagine that instructors will choose between skimping on examples or on existence and fixed-point theory. For what it is worth, if the choice must be made, I vote for doing a complete job on the examples, meaning Problems 14.2 through 14.5 and a full discussion of the Edgeworth Box. The level of abstraction in the definitions and theorems is, of course, ferocious, and I've found it really helps students to understand what a Walrasian equilibrium can look like, to work through the sort of parametric examples in the problems. Of course, to do this and to cover this chapter in one session means skimping on existence and fixed point theorems; that's not a great outcome, either. So, in the end, I vote for three sessions to cover this chapter and the next one, which is also fully packed. Even at three sessions, it will be a challenge to get through these two chapters.

Notes on the Problems

As I just indicated, I recommend doing at least a selection of the parametric examples given in Problems 14.2 through 14.5. Problem 14.1 provides good finger exercises concerning Edgeworth Boxes. Of the later, more abstract problems, Problems 14.8 and 14.9 are recommended for helping students internalize the use of Brouwer and Kakutani. Problem 14.10 (and a discussion of Mas-Colell's Theorem) is quite accessible and useful for hammering home the power of the S–M–D Theorem. Problem 14.11, which asks for a proof of Proposition 14.14 (upper semi-continuity of the Walrasian equilibrium correspondence in consumer endowments, assuming that equilibrium prices are nonzero and nonegative) is good review for techniques associated with (and extending) Berge's Theorem. My general rule is to include in the *Student's Guide* the proofs of in-the-text propositions that might prove at all difficult, but I made an exception in this case, because this is good "exercise" for students.

Solutions to Non-Starred Problems

■ 4. This is just like Problem 14.3, except that we begin with positive endowments of the two consumption goods, 1 and 2. The arguments that all four prices are strictly positive still hold. But we cannot be sure *apriori* that both firms are producing and, therefore, that equilibrium prices (normalized so that $p_3 = 1$) will be $p_1 = 1/3$ and $p_2 = 1/4$.

At first blush, four scenarios present themselves:

- $p_1 = 1/3$ and $p_2 = 1/4$, in which case firms can supplement (by their production) the supply of goods 1 and 2.
- $p_1 < 1/3$ and $p_2 = 1/4$, and only firm 2 produces output (good 2).
- $p_1 = 1/3$ and $p_2 < 1/4$, and only firm 1 produces output (good 1).
- $p_1 < 1/3$ and $p_2 < 1/4$, and neither firm produces output.

But the fourth of these is not, in fact, a possible equilibrium: If neither firm produces, there is no demand for good 3. But there is certainly supply of good 3 by Alice and Bob. And from Proposition 14.4 part e, we know that, in an economy with nonnegative prices and locally insatiable consumers, demand must *equal* supply for any good with a strictly positive price.

In fact, we know immediately that in the second scenario, all 10 units of good 3 must be purchased by firm 2 (supply *equals* demand again), which will turn this into 40 units of good 2. And in the third scenario, all 10 units of good 3 must be purchased by firm 1 and turned into 30 units of good 1.

Now it is a matter of checking each possible scenario. In scenario 1, you know all the prices, so you know the wealth levels of Alice and Bob, and you can work out demands at those prices. If this causes markets in good 3 to clear (if there is enough demand of the two consumption goods so that firms demand 10 units of good 3), there is an equilibrium of this type.

And in scenarios 2 and 3, while you don't know prices (you know one price but not the other), so you don't know wealth levels, you do know supplies of the two consumption goods, so you can check if the prices that cause those two output markets to clear (assuming the market in good 3 clears as just outlined) are consistent with the scenario.

If you do the work, you'll find that, for these numbers, it is (only) the first scenario that works out. The consumption bundles are $x^A = (19/2, 19)$ and $x_B = (95/8, 95/6)$, and the two production plans are $z^1 = (91/8, 0, -91/24)$ and $z^2 = (0, 149/6, -149/24)$.

(Unless you have reason to believe that this economy has a unique Walrasian equilibrium [there are theories about such things that do apply in this case, BUT the text did not discuss them], you have to check whether there aren't equilibria along the lines of scenario 2 or 3. Let me give you a bit of the calculations that show that, for instance, scenario 3 doesn't work: In scenario 3, we have 10 units of good 2 and 40 units of good 1 to be consumed. The wealth of each consumer is $5(1 + p_1 + p_2)$, where we know that $p_1 = 1/3$ and $p_2 < 1/4$. Call this number W; demand for good 1 depends (in this parametrization) on p_1 and W, and if total demand must be 40, you can work out that W will have to be $200/13.5$. But this then means that $p_2 = 22/13.5$, which is a good deal larger than $1/4$.)

■ 5. In this problem, the technologies have decreasing returns to scale, so the techniques applied to solve the previous two problems will not work. But you do know that the price all three commodities must be strictly positive, by the same logic as in previous problems. So normalize $p_3 = 1$.

Given p_1, firm 1 solves the profit-maximization problem

$$\text{maximize } p_1(3z_3^{1/3}) - z_3,$$

so the first-order condition is $p_1 z_3^{-2/3} = 1$, or $z_3 = p_1^{3/2}$, giving a supply of $3p_1^{1/2}$ and a profit of $p_1(3p_1^{1/2}) - p_1^{3/2} = 2p_1^{3/2}$.

Given p_2, firm 2 solves the profit-maximization problem

$$\text{maximize } p_2(2z_3^{1/2}) - z_3,$$

so the first-order condition is $p_2 z_3^{-1/2} = 1$, or $z_3 = p_2^2$, giving a supply of $2p_2$ and a profit of $p_2(2p_2) - p_2^2 = p_2^2$.

This means that at the prices p_1 and p_2 (and $p_3 = 1$), Alice's wealth is $5 + 2p_1^{3/2}$, while Bob's wealth is $5 + p_2^2$. Solving for their levels of demand, this yields

$$x_1^A = \frac{0.4(5 + 2p_1^{3/2})}{p_1} = \frac{2 + 0.8p_1^{3/2}}{p_1}, x_2^A = \frac{0.6(5 + 2p_1^{3/2})}{p_2} = \frac{3 + 1.2p_1^{3/2}}{p_2},$$

$$x_1^B = \frac{2.5 + 0.5p_2^2}{p_1}, \text{ and } x_2^B = \frac{2.5 + 0.5p_2^2}{p_2}.$$

Hence, we have an equilibrium when

$$\frac{2 + 0.8p_1^{3/2}}{p_1} + \frac{2.5 + 0.5p_2^2}{p_1} = 3p_1^{1/2}, \text{ and } \frac{3 + 1.2p_1^{3/2}}{p_2} + \frac{2.5 + 0.5p_2^2}{p_2} = 2p_2.$$

These are

$$4.5 + 0.8p_1^{3/2} + 0.5p_2^2 = 3p_1^{3/2} \text{ and } 5.5 + 1.2p_1^{3/2} + 0.5p_2^2 = 2p_2^2.$$

This gives

$$p_2 = 2.54587539 \quad \text{and} \quad p_1 = 2.31334227,$$

from which all the quantities can be computed.

■ 7. If, in building the generalized game, we use $p \cdot z^f$ in the consumer's budget constraint instead of $\pi^f(p)$, we encounter difficulties in demonstrating that the consumer's constraint correspondence is nonempty valued and continuous: Recall from the proof that we assumed endowments were strictly positive and firms could produce zero. The latter ensures that $\pi^f \geq 0$ at all prices, which, combined with the former, ensures that the consumer always has strictly positive wealth. With nonnegative wealth, we have nonemptiness of the constraint set; we needed strictly positive wealth (bounded away from zero, for normalized prices) for continuity. Using $p \cdot z^f$ means that firm f, with a particularly poor choice of z^f for the given prices p, could screw this up.

■ 10. It is evident that ζ as defined in the problem statement is a continuous function of p as long as p is in the interior of P. For every such p,

$$p \cdot \zeta(p) = d(p, Q)\left[\sum_{i=1}^{k} p_i\left(\frac{\hat{q}_i}{p_i} - 1\right)\right] = d(p, Q)\left[\sum_{i=1}^{k}(\hat{q}_i - p_i)\right] = 0,$$

since $\sum_i \hat{q}_i = \sum_i p_i = 1$ for all p (including $\hat{q}$) $\in P$. And the zeros of $\zeta(p)$ are precisely points $p \in Q$: For any $p \in Q$, $d(p, Q) = 0$, so the first term in the product zeroes out the second term. While if $p \notin Q$, then $d(p, Q) > 0$ and, since $p \neq \hat{q}$, for at least one component (say, component i), $p_i \neq \hat{q}_i$, so for that component, the second term in the product is not zero, hence the product is not zero.

Although you were told not to worry about it, it is worth pointing out that $\zeta_i(p)$ does explode to ∞ if p approaches a boundary of P along which the ith component of the price vector is zero.

■ 11. Suppose that $\{e_\ell\}$ is a sequence of endowment vectors converging to endowment vector e. (To clarify, the subscript ℓ here is for the sequence of endowment vectors. For consumer h's endowment in the ℓth economy, I'll write e_ℓ^h. I'll do my very best to avoid having to write consumer h's endowment of good i in the ℓth economy, as I've run out of locations to put the i.)

And suppose that, for each ℓ, $(p_\ell, \boldsymbol{x}_\ell, \boldsymbol{z}_\ell)$ is a Walrasian equilibrium for $\mathcal{E}(e_\ell)$, where $\lim_\ell (p_\ell, \boldsymbol{x}_\ell, \boldsymbol{z}_\ell)$ converges to $(p, \boldsymbol{x}, \boldsymbol{z})$. To show that the equilibrium correspondence is upper semi-continuous, we have to show that $(p, \boldsymbol{x}, \boldsymbol{z})$ is a Walrasian equilibrium for $\mathcal{E}(e)$.

Market clearing is easy: $\sum_h \boldsymbol{x}_\ell^h \leq \sum_h e_\ell^h + \sum_f \boldsymbol{z}_\ell^f$ for each ℓ, and we can pass to the limit and get the same inequality without the ℓs.

For the profit-maximization condition, fix a firm f. Since (by assumption) Z^f is closed, $\boldsymbol{z}^f = \lim_\ell \boldsymbol{z}_\ell^f$ is feasible for the firm. And to show that it is profit maximizing, take any other $\hat{z} \in Z^f$. Since $\boldsymbol{z}_\ell^f$ is profit maximizing for f at prices p_ℓ, we have

$$p_\ell \cdot \boldsymbol{z}_\ell^f \geq p_\ell \cdot \hat{z}.$$

(Note that $\hat{z}$ is feasible for f in the economies "along the sequence," since Z^f doesn't change.) Passing to the limit on either side of this inequality shows that $p \cdot \boldsymbol{z}^f \geq p \cdot \hat{z}$, and since this is true for all $\hat{z} \in Z^f, \boldsymbol{z}^f$ must be profit maximizing at the prices p.

And for utility maximization, fix a consumer h and write $y^h = p \cdot e^h + \sum_f s^{fh} p \cdot \boldsymbol{z}^f$, and also, for each ℓ, $y_\ell^h = p_\ell \cdot e_\ell^h + \sum_f s^{fh} p_\ell \cdot \boldsymbol{z}_\ell^f$. Because $p_\ell \to p$, $e_\ell^h \to e^h$, and $\boldsymbol{z}_\ell \to \boldsymbol{z}$, we know that $\lim_\ell y_\ell^h = y^h$. Now fix any bundle $\hat{x}$ that is budget feasible in the limit situation; that is, that satisfies

$$p \cdot \hat{x} \leq p \cdot e^h + \sum_f s^{fh} p \cdot \boldsymbol{z}^f = y^h.$$

Consider three cases:

In the first case, if $y^h = 0$, then $\hat{x} = \boldsymbol{x}^h = 0$ because both are budget feasible (prices p are nonnegative and the consumer's consumption space is R_+^k), and so $u^h(\hat{x}) = u^h(\boldsymbol{x}^h)$.

Case 2 is $y^h < 0$. Then for sufficiently large ℓ, $y_\ell^h < 0$. But then $(p_\ell, \boldsymbol{x}_\ell, \boldsymbol{z}_\ell)$ cannot be a Walrasian equilibrium for these large ℓ, as consumer h cannot find a feasible consumption bundle that she can afford (once again, because prices are nonnegative and $X^h = R_+^k$). So $y^h < 0$ is impossible.

This leaves $y^h > 0$. Since $\lim_\ell y_\ell^h = y^h$, we know that, for all sufficiently large ℓ, $y_\ell^h > 0$. Consider two subcases: First, if $p \cdot \hat{x} = 0$, then $\hat{x} = 0$, and so $\hat{x}$ is affordable at prices p_ℓ for all sufficiently large ℓ. But then $u^h(\hat{x}) \leq u^h(\boldsymbol{x}_\ell^h)$ and, by the continuity of u^h, $u^h(\hat{x}) \leq u^h(\boldsymbol{x}^h)$.

In the second subcase, $p \cdot \hat{x} > 0$, hence for all large ℓ, $p_\ell \cdot \hat{x} > 0$. For large enough ℓ so this is true, let

$$\alpha_\ell = \frac{y_\ell^h}{y^h} \times \frac{p \cdot \hat{x}}{p_\ell \cdot \hat{x}}.$$

Because $\lim_\ell y^h_\ell = y^h$ and $\lim_\ell p_\ell = p$, we know that $\lim_\ell \alpha_\ell = 1$. Now

$$p_\ell \cdot (\alpha_\ell \hat{x}) = \alpha_\ell (p_\ell \cdot \hat{x}) = \frac{y^h_\ell}{y^h} \times \frac{p \cdot \hat{x}}{p_\ell \cdot \hat{x}} \times p_\ell \cdot \hat{x} = y^h_\ell \times \frac{p \cdot \hat{x}}{y^h} \leq y^h_\ell.$$

Therefore, $\alpha_\ell \hat{x}$ is affordable by h at prices p_ℓ. But since $\boldsymbol{x}^h_\ell$ is optimal for h at these prices, $u^h(\alpha_\ell \hat{x}) \leq u^h(\boldsymbol{x}^h_\ell)$. Passing to the limit on both sides of this inequality, using the continuity of u^h, shows that $u^h(\hat{x}) \leq u^h(\boldsymbol{x}^h)$. Done. ■